

EL CALAFATE LAGO ARGENTINO, Argentina

WMO: 879030

Lat: 50.336S

Lon: 72.249W

Elev: 722

StdP: 14.32

Time Zone: -3.00 (W03)

Period: 91-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	20.2	22.5	8.1	8.6	47.4	11.0	9.9	45.7	26.6	41.1	23.3	38.7	2.0	N/A	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.0	72.0	53.4	69.2	51.3	66.6	50.0	55.3	68.6	53.3	65.6	51.8	63.7	12.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
49.1	53.0	57.8	46.5	48.0	57.3	44.4	44.3	56.5	23.7	69.3	22.5	65.6	21.5	63.8	66.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
32.6	27.3	23.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB	12.7	60.8	2.6	2.7	10.8	62.7	9.3	64.3	7.9	65.9	6.0	67.8
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.6	56.0	55.0	52.0	47.0	40.1	32.9	33.2	37.5	41.3	47.2	51.6	53.9
	DBStd	9.64	4.76	5.32	4.96	5.32	5.53	5.61	6.44	5.30	5.67	5.22	4.78	4.91
	HDD50	2337	5	12	35	120	309	512	520	386	265	117	38	18
	HDD65	7088	279	281	402	541	773	962	985	851	711	553	403	347
	CDD50	730	193	153	98	30	1	0	0	0	4	29	85	137
	CDD65	4	2	1	0	0	0	0	0	0	0	0	0	1
	CDH74	26	7	10	2	0	0	0	0	0	0	0	1	6
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.6	11.7	9.6	6.6	5.3	4.1	2.9	4.8	6.7	7.8	9.9	10.9	11.5
Precipitation	PrecAvg	7.9	0.5	0.4	0.6	1.0	1.2	0.9	1.1	0.8	0.7	0.5	0.4	0.5
	PrecMax	14.3	2.3	1.3	1.6	3.2	3.0	3.1	3.0	1.9	1.7	1.5	1.4	3.2
	PrecMin	2.3	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	3.4	0.5	0.3	0.5	0.8	0.8	0.7	0.7	0.5	0.5	0.4	0.4	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.9	75.4	72.6	65.5	57.8	50.4	52.9	54.9	61.2	65.5	70.0	74.5
		MCWB	54.7	56.4	54.7	50.6	45.6	41.4	42.1	41.4	45.1	47.8	50.5	54.9
	2%	DB	71.0	71.5	67.6	61.3	53.3	46.8	49.4	51.5	56.2	62.0	66.5	69.9
		MCWB	52.3	54.0	51.3	48.4	43.4	39.0	40.0	40.9	42.2	46.0	49.2	50.9
	5%	DB	68.3	68.1	64.2	58.9	50.6	44.0	46.1	48.7	53.4	59.4	64.0	66.4
		MCWB	50.5	52.0	49.5	46.9	41.6	37.0	38.0	38.9	41.2	44.5	47.9	49.8
	10%	DB	65.6	64.7	61.4	55.9	48.1	41.1	42.8	46.2	50.8	57.2	61.6	63.7
		MCWB	49.0	49.6	48.1	44.9	40.3	35.3	36.2	37.4	39.8	44.0	46.3	47.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.2	58.2	55.9	54.7	47.0	42.3	43.4	44.0	45.7	50.5	55.3	56.8
		MCDB	71.5	71.1	68.9	61.5	54.3	48.7	51.0	52.2	58.0	60.7	65.5	70.5
	2%	WB	53.7	55.0	53.3	49.7	44.6	39.7	40.5	41.3	43.5	47.7	51.2	53.6
		MCDB	67.6	68.8	63.2	59.4	50.8	45.9	47.8	49.5	54.4	58.3	62.8	66.8
	5%	WB	52.1	52.7	51.4	47.6	42.9	37.6	38.6	39.7	41.9	46.2	49.2	51.2
		MCDB	65.8	65.3	61.1	56.4	48.7	42.9	45.0	47.3	51.9	56.8	61.5	63.3
	10%	WB	50.3	50.9	49.7	45.9	41.4	35.9	36.7	38.1	40.3	44.7	47.7	49.3
		MCDB	64.0	62.9	59.5	54.4	46.8	40.4	42.6	45.3	49.5	55.7	59.6	61.0
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.0	15.3	14.1	11.4	9.9	10.2	12.0	13.8	15.9	17.0	16.2
		MCDBR	19.5	19.4	18.8	17.4	14.7	13.0	13.8	15.3	17.6	19.1	20.2	19.5
		MCWBR	10.1	10.4	10.6	10.7	9.2	8.5	9.1	8.9	9.9	10.4	11.5	10.3
	5% WB	MCDBR	17.7	17.9	16.4	15.7	13.3	12.2	13.3	14.6	16.8	17.1	18.5	16.9
		MCWBR	10.3	10.7	10.3	11.3	9.3	8.6	9.3	9.6	10.1	10.7	12.3	10.9
Clear-Sky Solar Irradiance	taub	0.308	0.306	0.293	0.281	0.259	0.245	0.249	0.266	0.287	0.299	0.300	0.310	
	taud	2.553	2.569	2.599	2.595	2.525	2.397	2.385	2.419	2.463	2.517	2.542	2.527	
	Ebn at Noon	318	309	296	271	245	227	241	269	291	308	319	320	
	Edh at Noon	32	30	25	20	16	16	18	23	28	31	32	34	
All-Sky Solar Radiation	RadAvg	1848	1557	1078	653	376	262	322	555	946	1349	1734	1895	
	RadStd	117	88	60	41	35	21	28	56	74	79	97	119	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page